

Normative Brain TSPO-PET Mapping Across Healthy Aging

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Status: Portfolio project — independent analysis, not affiliated with the original data collection.

Motivation

My MSc thesis examined normative ranges for hippocampal volume from structural MRI in a healthy population. I built an automated FSL pipeline for it (FIRST segmentation, intracranial-volume correction, group comparisons).

This project asks a similar question with a different modality. Using only the healthy-control subgroup of a public OpenNeuro dataset, I built a small pipeline that:

1. registers each subject's T1-weighted MRI to standard (MNI) space,
2. carries that transform over to their PET image,
3. extracts regional PET signal, in SUV (g/mL), from standard anatomical ROIs, and
4. checks how that regional signal relates to age in a healthy population.

So instead of "what is the normal range of hippocampal volume," the question here is "what is normal regional TSPO-PET signal, and how does it shift with age." Same logic, applied to molecular PET instead of volumetrics.

This is meant to show a working PET analysis pipeline (registration, atlas-based quantification, SUV computation, basic normative statistics). It uses only the healthy-control subgroup; no patient or clinical data, no group comparisons.

Data

The data are obtained from OpenNeuro dataset ds007907, titled "*Elevated translocator protein in skull bone marrow is linked to human chronic pain*" (Mohammadian, Loggia et al.). The dataset is publicly available under a CC0 license.

Only the healthy-control (HC) participants are included in the current analysis. The HC group consists of approximately 20–25 participants, with ages ranging from young adulthood to older adulthood.

1. **PET:** Static [^{11}C]PBR28 PET images are used as a marker of TSPO expression and neuroinflammation. The images represent the 60–90 minute post-injection period and are provided in SUV units (g/mL).

2. **MRI:** T1-weighted MPRAGE structural MRI scans acquired at 3T are used for anatomical processing and registration. The MRI images were already co-registered to the PET images by the original data providers.
3. **Data selection:** Raw imaging data are not stored in this repository. Instead, the script `01_get_hc_subjects.sh` downloads only the healthy-control subjects required for the analysis directly from the public OpenNeuro S3 bucket.

Pipeline

- **Get Healthy Control Subjects:** Download only healthy control subjects (`sub-HC*`) including T1-weighted MRI, PET (SUV), and demographic metadata.
- **Preprocess T1 and PET:** Perform brain extraction using BET, followed by linear and nonlinear registration to MNI152 2 mm space. Apply the same transformation to the PET images while preserving PET values in SUV (g/mL).
- **Extract ROI Values:** Calculate the mean PET signal within the selected Harvard-Oxford atlas ROIs using Nilearn, and merge the extracted values with participants' age and sex information. The final measure is reported as SUV (g/mL).
- **Statistical Analysis and Visualization:** Perform Pearson correlation and partial correlation between age and PET SUV while controlling for sex. Generate scatter plots and a summary bar chart, with PET values reported in SUV (g/mL) and age in years.

Requirements

1. **FSL:** FSL version 6.0 or later is required for the neuroimaging preprocessing and registration steps. The pipeline uses `bet`, `flirt`, `fnirt`, and `applywarp`.
2. **Neurodesk Play:** If FSL is not installed locally, the entire pipeline can be run through Neurodesk Play, which provides a browser-based Linux environment without requiring a local installation. After opening the Neurodesk desktop, open a terminal and run `ml fsl` to load FSL.
3. **Python packages:** The required Python dependencies can be installed using the provided `requirements.txt` file by running `pip install -r requirements.txt`.
4. **AWS CLI:** The data download script also requires `awscli`. It can be installed using `pip install awscli`.

How to Run

1. **Download the HC data:** Run `01_get_hc_subjects.sh` to download the healthy-control PET, MRI, and metadata files. The downloaded data will be saved in the `data/` directory.
2. **Preprocess MRI and PET:** Run `02_preprocess_t1_pet.sh` to perform brain extraction and image registration. This step requires FSL to be installed and available on the system.
3. **Extract ROI values:** Run `03_extract_roi_values.py` to calculate the mean PET signal within the selected ROIs. The extracted values will be saved as `results/roi_suv_values.csv`.
4. **Perform statistical analysis:** Run `04_analyze_and_plot.py` to perform the correlation analyses and generate the corresponding figures. The statistical results and plots will be saved in the `results/` directory.

Limitations

1. **No validated reference region:** [^{11}C]PBR28 does not have a brain region that can be considered free of specific TSPO binding. Therefore, the analysis uses raw SUV values. Any SUVR analysis would require a conventional reference region rather than a validated kinetic reference.
2. **No genotype correction:** PBR28 binding affinity is influenced by the TSPO rs6971 polymorphism. Since genotype information is not available in the public participant table, this potential source of variability cannot be accounted for in the current analysis.
3. **Small, cross-sectional sample:** The study includes a relatively small number of participants, with only one scan per subject. The sample is sufficient to demonstrate the analysis pipeline, but it is not large enough to support strong conclusions about age-related changes in TSPO binding.
4. **Atlas-based parcellation:** ROIs are defined using a standard-space atlas after nonlinear registration rather than subject-specific anatomical segmentation. This approach simplifies the analysis but may reduce anatomical precision compared with individualized ROI delineation.

Citation

Mohammadian M, Efthimiou N, Gabriel K, Brusafferri L, Cooper-Hohn J, Kim M, Murphy JP, Alshelh Z, Grmek G, Schnieders JH, Chane CA, Carmichael TG, Yang D, Schubert JJ, Turkheimer FE, Cazuza RA, Grace PM, Catana C, Stufflebeam SM, Edwards RR, Napadow V, Sullivan K, Nahrendorf M, Price TJ, Gilman JM, Loggia ML. *Evidence for a role of skull bone marrow in human chronic pain (as revealed by TSPO PET imaging)*. Science Translational Medicine (in press). doi: 10.1126/scitranslmed.aed8729.

Preprint: doi: 10.1101/2025.07.19.25331817

Dataset: OpenNeuro ds007907. doi: 10.18112/openneuro.ds007907.v1.0.2

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